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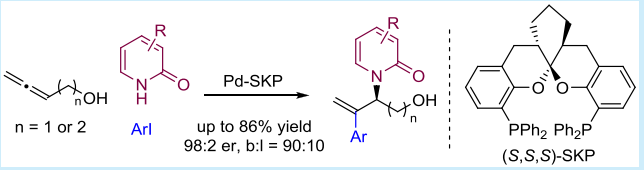
Pd-Catalyzed Regio- and Enantioselective Aminoarylation of Allenols with Aryl Iodides and 2-Pyridones

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ABSTRACT: A new asymmetric catalytic protocol for the synthesis of enantioenriched *N*-allyl 2-pyrodones has been developed via the first Pd-catalyzed regio- and enantioselective aminoarylation of allenols with aryl iodides and 2-pyridones. By using a palladium complex generated in situ from $Pd_2(dba)_3 \cdot CHCl_3$ and (*S*,*S*,*S*)-SKP as a catalyst, the three-component aminoarylation proceeded smoothly to afford a variety of functionalized *N*-allylic 2-pyridones in high yields with good regioselectivities and excellent enantioselectivities.



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Chiral *N*-substituted 2-pyridones have emerged as useful motifs in many natural products and medicinally relevant agents, as exemplified by aspernigrin B[1] isolated from *Axinella damicornis* and *Aspergillus welwitschiae* CUGBMF180262, respectively, human rhinovirus 3C protease inhibitors,[2] GDC-0994 as an extracellular-signal-regulated kinase inhibitor[3] (Figure 1), and many others.[4] In this context, the development

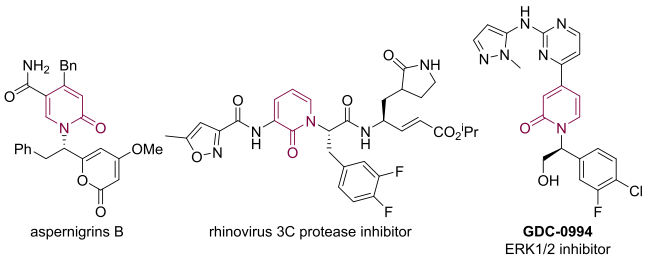
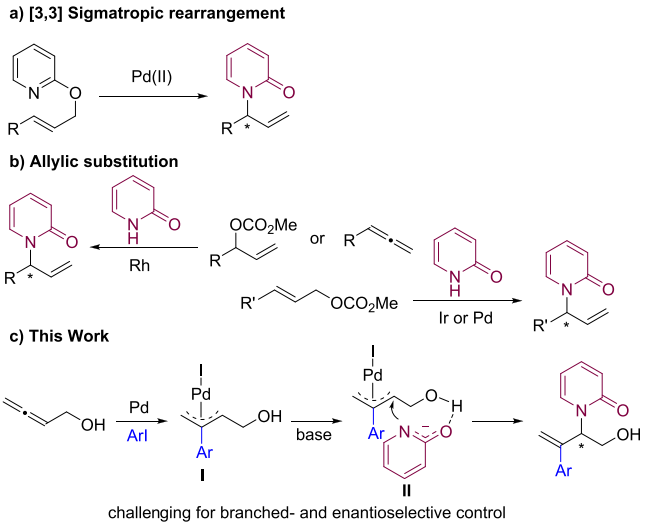


Figure 1. Examples of bioactive chiral *N*-substituted 2-pyridones.

of synthetic approaches to chiral *N*-substituted 2-pyridone derivatives became an important goal in pharmaceutical research and drug discovery. In general, they are prepared via stereoselective nucleophilic substitution of 2-pyridones to chiral electrophiles, or by the reaction of 2*H*-pyran-2-ones with chiral amines.[2−5] However, asymmetric catalytic protocols for the synthesis of chiral *N*-substituted 2-pyridones from achiral starting materials are less explored. In 2010, Batey and co-workers reported first asymmetric catalytic method to access enantioenriched N-substituted 2-pyridones via Pd(II)-catalyzed asymmetric [3,3] sigmatropic rearrangement of 2-allyloxy pyridines (Scheme 1a).[6] So far, only sporadic reports on asymmetric catalytic approaches have been documented, which are restricted to the allylic substitution of allylic donors with 2-pyridones catalyzed by a chiral Rh-,[7] Ir-,[8] or Pd-complex[9] (Scheme 1b). Therefore, the development of efficient asymmetric catalytic methods to provide more functionalized

Scheme 1. Asymmetric Catalytic Approaches to Chiral *N*-Substituted Pyridones



chiral N-substituted 2-pyridones from readily accessible starting materials is highly appealing.

Since the seminal work by Tsuji and Shimizu in 1984 on the Pd-catalyzed aminoarylation of allenes,[10] Pd-catalyzed carbofunctionalization of allenes with aryl halides and nucleophiles has become a useful tool for the preparation of valuable compounds with diverse functionalities.[11] In contrast to the

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Pd-catalyzed allylic alkylation of allylic substrates,[12] the three-component reaction can avoid preinstallation of a leaving group for the allylic donors, and allows to construct two chemical bonds in one-pot manipulation.[13] However, the Pd-catalyzed carbofunctionalization of allenes generally gives linear allylic products, whereas controlling the regioselectivity to afford branched products remains elusive.[14] Furthermore, it is important to point out that the asymmetric versions of this type of carbofunctionalization of allenes have not yet been documented so far to the best of our knowledge.

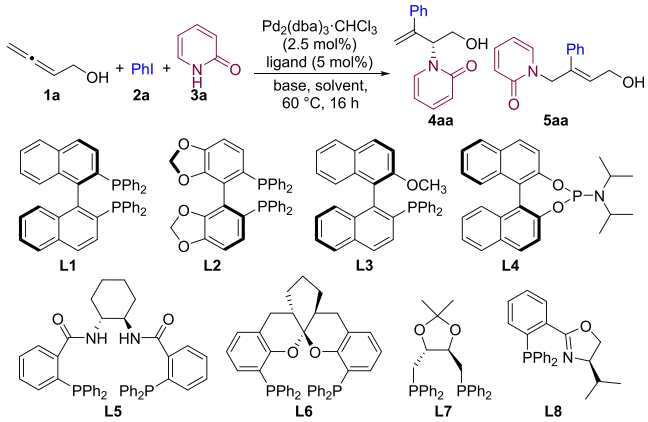
As an ongoing effort on the development of Pd-catalyzed regio- and enantioselective allylic substitution[15] and cycloaddition,[16] we are interested to develop a new asymmetric catalytic protocol for the construction of enantioenriched *N*-substituted 2-pyridones via Pd-catalyzed three-component aminoarylation of allenols with aryl iodides and 2-pyridones. We envisioned that the Heck-type arylation of allenol with aryl iodides could give 2-aryl substituted allylpalladium intermediate **I**, which might undergo allylic substitution with 2-pyridone to produce desired *N*-substituted pyridones (Scheme 1c). Herein, we report the first Pd-catalyzed regio- and enantioselective three-component process of carbofunctionalization of allenes, which allows one to rapidly access more functionalized *N*-substituted 2-pyridones in high yields with good regioselectivities and excellent enantioselectivities via the formation two chemical bonds from readily accessible starting materials.

Initial investigation was focused on the optimization of the conditions for the Pd-catalyzed aminoarylation using 2,3-allenol (**1a**), iodobenzene (**2a**), and 2-pyrodone (**3a**) as standard substrates (Table 1). The reactions were typically performed in the presence of K_2CO_3 as the base additive in toluene at 60 °C for 16 h, with the palladium complex generated in situ from $Pd_2(dba)_3{\cdot}CHCl_3$ and a chiral ligand as the catalyst. At the outset, some chiral ligands with different skeletons and/or ligating moieties were examined (entries 1−8). To our delight, (*S*,*S*,*S*)-SKP (**L6**)[17] proved to be the optimal chiral ligand, to afford the targeting branched-product **4aa** in 73% yield with 93:7 er and a b/l ratio of 80:20 (entry 6). With (*S*,*S*,*S*)-SKP (**L6**) as a ligand, reducing the loading of K_2CO_3 to 1 equiv led to an increment of er to 98:2 (entry 9). A brief survey of the solvent effect uncovered toluene as the optimal solvent (entries 9−13). The reactions proceeded much better using K_2CO_3 or K_3PO_4 (entries 9 or 15) as the base to afford **4aa** with higher enantioselectivities, demonstrating significant impact of base on the reaction efficiency (entries 14−18 and 9).

With optimized conditions in hand (Table 1, entry 9), the reaction generality was then investigated by the three component aminoarylation of 2,3-allenol (**1a**) with 2-pyridone (**3a**) and various aryl iodides **2**. As shown in Figure 2, a wide range of aryl iodides bearing different steric and electronic features tolerates the reaction conditions very well to afford the corresponding carboaminated products **4aa**−**4ja** in moderate to high yields with good branched regioselectivities and excellent enantioselectivities. While the reaction of 2-naphthyl iodide proceeded smoothly to give the corresponding product **4ka** in high yield with excellent enantioselectivity, a significant drop of efficiency was observed when 1-naphthyl iodide was employed in the reaction.

The substrate scope for aryl iodides **2** and substituted 2-pyridones **3** was subsequently studied in the reactions of 2,3-allenol **1a** (Figure 3). Under the optimized conditions A

Table 1. Optimization of Conditions[a]



entry	ligand	base (eq)	solvent	yield (%)[b]	er[c]	4aa:5aa[d]
1	L1	K_2CO_3 (2)	toluene	10	85:15	71:29
2	L2	K_2CO_3 (2)	toluene	48	73:27	75:25
3	L3	K_2CO_3 (2)	toluene	30	90:10	74:26
4	L4	K_2CO_3 (2)	toluene	NR		
5	L5	K_2CO_3 (2)	toluene	50	77:23	67:33
6	L6	K_2CO_3 (2)	toluene	73	93:7	80:20
7	L7	K_2CO_3 (2)	toluene	NR		
8	L8	K_2CO_3 (2)	toluene	NR		
9	L6	K_2CO_3 (1)	toluene	70	98:2	79:21
10	L6	K_2CO_3 (1)	THF	59	81:19	67:33
11	L6	K_2CO_3 (1)	CPME	67	92:8	74:26
12	L6	K_2CO_3 (1)	CH_2Cl_2	45	70:30	57:43
13	L6	K_2CO_3 (1)	CH_3CN	60	88:12	71:29
14	L6	Cs_2CO_3 (1)	toluene	68	96:4	78:22
15	L6	K_3PO_4 (1)	toluene	60	97:3	74:26
16	L6	KOH (1)	toluene	73	73:27	76:24
17	L6	KF (1)	toluene	30	93:7	60:40
18	L6	Et_3N (1)	toluene	42	93:7	60:40

[a]Reaction conditions: $Pd_2(dba)_3{\cdot}CHCl_3$ (2.5 mol %), ligand (5 mol %), base (0.2 or 0.4 mmol), **1a** (0.2 mmol), **2a** (0.24 mmol), **3a** (0.24 mmol), solvent (2.0 mL), 60 °C, 16 h. [b]The yield of isolated **4aa**. [c]Determined by HPLC on a chiral stationary phase. The absolute configuration was assigned by X-ray crystallography for **4aa** (see the Supporting Information). [d]Determined by ^1H NMR spectroscopy of the crude reaction mixture.

(Table 1, entry 9), the reaction with 5-methyl-2-pyridone proceeded smoothly to furnish aminoarylated products **4ab** in good yields with moderate b/l ratio and high enantioselectivity (98:2 er). However, moderate enantioselectivities were observed for the reactions with other substituted 2-pyridones. In this context, the reaction conditions were further optimized toward the reaction of **1a** with **2a** and 5-nitro-2-pyridone (**3f**) (see the Supporting Information for details). As a result, we found that the enantioselectivity could be improved in most cases when the reactions were conducted in the presence of K_3PO_4 as the base in methyl *tert*-butyl ether (MTBE) as a solvent at 30 °C for 24 h (condition B). This optimized condition B was compatible with a diversity of aryl iodides **2** or substituted 2-pyridones **3** bearing various functional groups, such as nitro, aldehyde, cyanide, bromide, and 2-naphthyl groups, to produce the corresponding adducts **4** in good yields with moderate b/l ratios and high enantioselectivities.

To gain insight into the mechanism for the present three-component aminoarylation process, several monosubstituted

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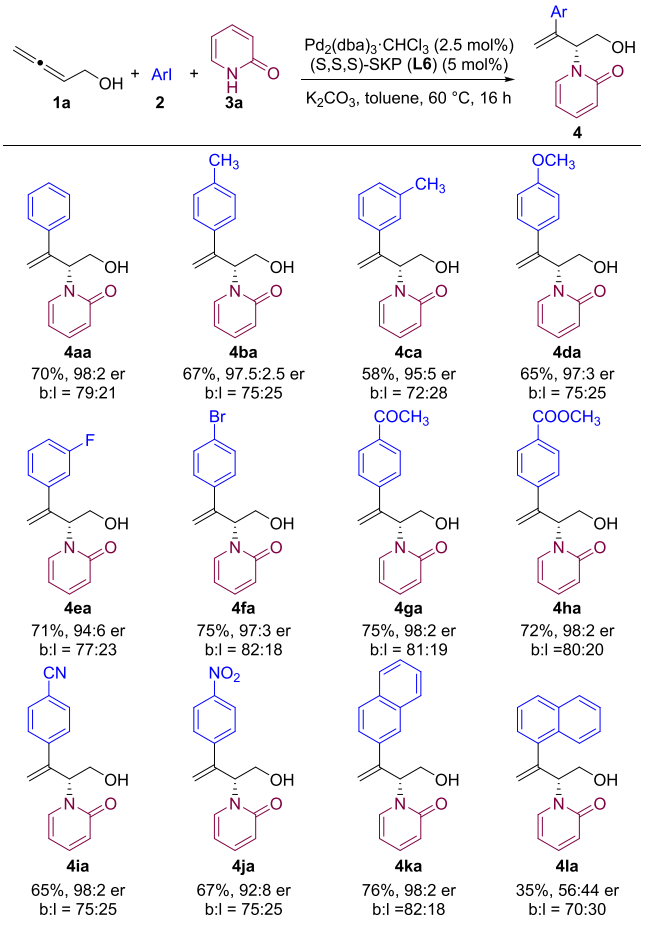


Figure 2. Pd-Catalyzed aminoarylation of 2,3-allenol (**1a**) with 2-pyridone (**3a**) and various aryl iodides **2**. Reaction conditions: $Pd_2(dba)_3{\cdot}CHCl_3$ (2.5 mol %), (*S*,*S*,*S*)-SKP (5 mol %), K_2CO_3 (0.2 mmol), **1a** (0.2 mmol), **2** (0.24 mmol), **3a** (0.24 mmol), toluene (2.0 mL), 60 °C, 16 h. The yields are of isolated materials. The enantioselectivities were determined by HPLC on a chiral stationary phase. The branched/linear ratios of products were determined by ^1H NMR spectroscopy of the crude reaction mixture.

allenes **1b**-**1e** were examined in the reaction with **2a** and **3a** under the optimized conditions **A** (Table 1, entry 9). As shown in Scheme 2, the reaction of 3,4-allenol **1b** with **2a** and **3a** proceeded smoothly to afford corresponding aminoarylated product **6a** in high yield with high branched selectivity and enantioselectivity. In sharp contrast, the reaction of 4,5-allenol **1c** with a longer distance between hydroxyl and allenyl moieties failed to produce the corresponding aminoarylated product under otherwise identical conditions. In addition, 2,3-allenol ether **1d** with the hydroxyl group being blocked by *n*-butyl and alkyl-substituted allene **1e** were also found incompatible with the protocol. These results implied that a properly situated hydroxyl on the monosubstituted allene is essential not only for the branched regioselectivity control, but also for the reactivity. The appropriate directing effect on hydroxyl group might be critical for the insertion step of allenols to Ar−Pd−I intermediates. In addition, the allylpalladium intermediates bearing a hydroxymethyl or hydroxyethyl groups might build an effective hydrogen-bond interaction with 2-pyridones,[18] which can improve the nucleophilicity of 2-pyridones on one hand, and possibly favors the branched regioselectivity on the other. As a further demonstration, the aminoarylation of 3,4-allenol **1b** with different aryl iodides and

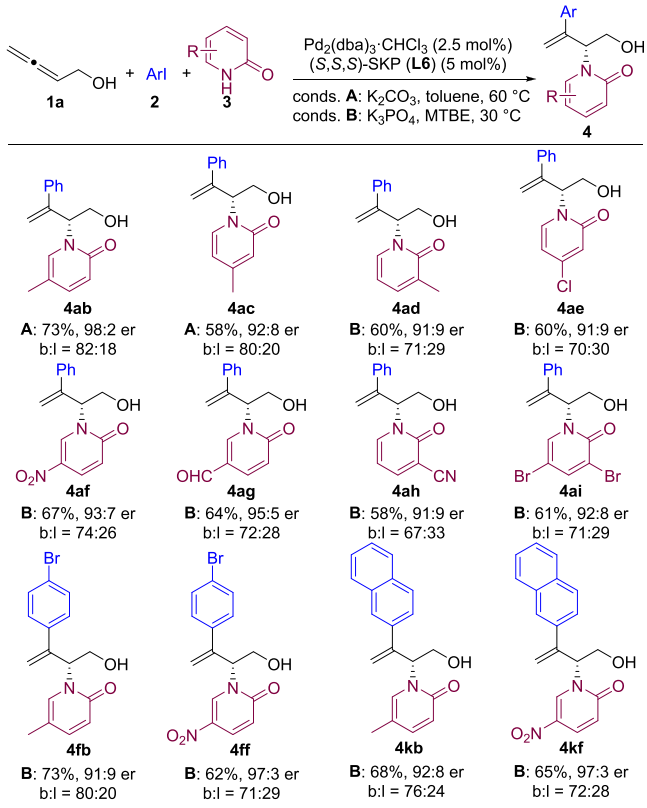
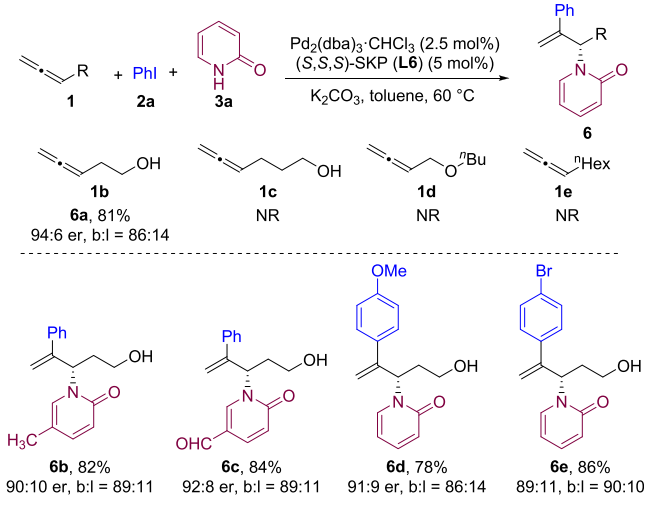


Figure 3. Pd-Catalyzed aminoarylation of 2,3-allenol (**1a**) with various aryl iodides **2** and 2-pyridones **3**. Reaction conditions A: As Figure 2. Reaction conditions B: $Pd_2(dba)_3{\cdot}CHCl_3$ (2.5 mol %), (*S*,*S*,*S*)-SKP (5 mol %), K_3PO_4 (0.2 mmol), **1a** (0.2 mmol), **2** (0.24 mmol), **3** (0.24 mmol), MTBE (2.0 mL), 30 °C, 24 h. The yields are of isolated materials. The enantioselectivities were determined by HPLC on a chiral stationary phase. The branched/linear ratios of products were determined by ^1H NMR spectroscopy of the crude reaction mixture.

Scheme 2. Pd-Catalyzed Amioarylation of Allenes 1 with 2 and 3



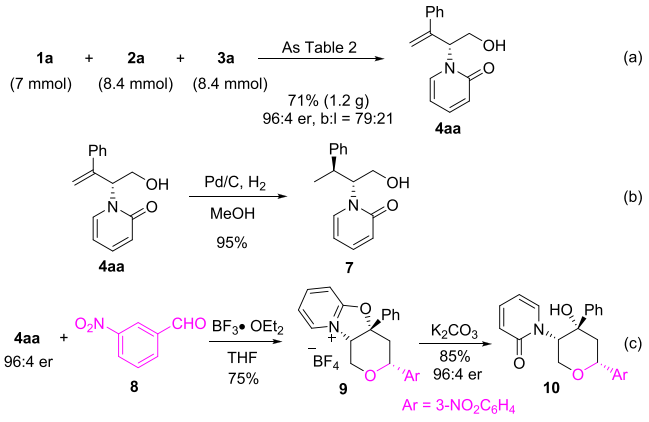
substituted 2-pyridones still proceeded well to afford the corresponding aminoarylated products **6** in high yields with high levels of regio- and enantioselectivities.

To demonstrate the synthetic utility of present process, the gram-scale transformation and the elaboration of the product

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have been investigated (Scheme 3). The three-component aminoarylation of **1a** with **2a** and **3a** in 7 mmol scale under

Scheme 3. Gram-Scale Transformation and the Elaboration of Product 4aa



standard conditions still proceeded smoothly to give desired product **4aa** in 71% yield (1.2 g) with a 79:21 b/l ratio and 96:4 er (Scheme 3a). With the obtained **4aa**, the further transformation has been examined. The selective hydrogenation of **4aa** in the presence of Pd/C as a catalyst in methanol afforded compound **7** as an only diastereomer in 95% yield (Scheme 3b). Next, the Prins-type cyclization[19] of **4aa** with 3-nitrobenzaldehyde (**8**) was carried out in the presence of BF_3 etherate in THF. Interestingly, the bicyclic pyridium salt **9** bearing three stereocenters can be obtained in 75% yield with complete diastereoselectivity. The hydrolysis[20] of compound **9** with aqueous K_2CO_3 proceeded smoothly to afford more functionalized chiral 2-pyridone compound **10** as an only diastereomer in 85% yield without any deterioration of enantiomeric purity (Scheme 3c). The absolute configurations of **7** and **9** were unambiguously assigned by X-ray crystallography (see the Supporting Information).

In conclusion, we have developed a new asymmetric catalytic protocol for the synthesis valuable *N*-substituted 2-pyridone derivatives via the first Pd-catalyzed enantioselective three-component aminoarylation of allenols with aryl iodides and 2-pyridones. By using palladium complex generated in situ from $Pd_2(dba)_3{\cdot}CHCl_3$ and (*S*,*S*,*S*)-SKP as a catalyst, the three-component aminoarylation proceeded smoothly through the formation of a carbon−carbon and a carbon−nitrogen bond in one pot, providing a variety of functionalized *N*-allylic 2-pyridones in high yields with good regioselectivities and excellent enantioselectivities. The control experiments demonstrated that the hydroxyl group on the allenols is essential for the reactivity of the process and the control of the branched selectivity. The synthetic utility of the present process has been demonstrated by gram-scale transformation and the elaboration of the product. Further investigation of this type of three-component tandem reaction is currently underway in our laboratory and will be reported in due course.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c00959.

Experimental details and compound characterization data (PDF)

Accession Codes

CCDC 2036007−2036008 and 2061837 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) (a) Hiort, J.; Maksimenka, K.; Reichert, M.; Perović-Ottstadt, S.; Lin, W. H.; Wray, V.; Steube, K.; Schaumann, K.; Weber, H.; Proksch, P.; Ebel, R.; Müller, W. E. G.; Bringmann, G. New natural products from the sponge-derived fungus *Aspergillus niger*. *J. Nat. Prod.* **2004**, *67*, 1532−1543. (b) Han, J.; Yang, N.; Wei, S.; Jia, J.; Lin, R.; Li, J.; Bi, H.; Song, F.; Xu, X. Dimeric hexylitaconic acids from the marine-derived fungus Aspergillus welwitschiae CUGBMF180262. *Nat. Prod. Res.* **2020**, 1.

(2) (a) Dragovich, P. S.; Prins, T. J.; Zhou, R.; Johnson, T. O.; Hua, Y.; Luu, H. T.; Sakata, S. K.; Brown, E. L.; Maldonado, F. C.; Tuntland, T.; Lee, C. A.; Fuhrman, S. A.; Zalman, L. S.; Patick, A. K.; Matthews, D. A.; Wu, E. Y.; Guo, M.; Borer, B. C.; Nayyar, N. K.; Moran, T.; Chen, L.; Rejto, P. A.; Rose, P. W.; Guzman, M. C.; Dovalsantos, E. Z.; Lee, S.; McGee, K.; Mohajeri, M.; Liese, A.; Tao, J.; Kosa, M. B.; Liu, B.; Batugo, M. R.; Gleeson, J.-P. R.; Wu, Z. P.; Liu, J.; Meador, J. W., III; Ferre, R. A. Structure-based design, synthesis, and biological evaluation of irreversible human rhinovirus 3C protease inhibitors. 8. Pharmacological optimization of orally bioavailable 2-pyridone-containing peptidomimetics. *J. Med. Chem.* **2003**, *46*, 4572−4585. (b) Dragovich, P. S.; Prins, T. J.; Zhou, R.; Brown, E. L.; Maldonado, F. C.; Fuhrman, S. A.; Zalman, L. S.; Tuntland, T.; Lee, C. A.; Patick, A. K.; Matthews, D. A.; Hendrickson, T. F.; Kosa, M. B.; Liu, B.; Batugo, M. R.; Gleeson, J. P. R.; Sakata, S. K.; Chen, L.; Guzman, M. C.; Meador, J. W., III; Ferre, R. A.; Worland, S. T. Structure-based design, synthesis, and biological evaluation of irreversible human rhinovirus 3C protease inhibitors. 6. Structure-activity studies of orally bioavailable, 2-pyridone-containing peptidomimetics. *J. Med. Chem.* **2002**, *45*, 1607−1623.

(3) Blake, J. F.; Burkard, M.; Chan, J.; Chen, H.; Chou, K.-J.; Diaz, D.; Dudley, D. A.; Gaudino, J. J.; Gould, S. E.; Grina, J.; Hunsaker, T.; Liu, L.; Martinson, M.; Moreno, D.; Mueller, L.; Orr, C.; Pacheco, P.; Qin, A.; Rasor, K.; Ren, L.; Robarge, K.; Shahidi-Latham, S.; Stults, J.; Sullivan, F.; Wang, W.; Yin, J.; Zhou, A.; Belvin, M.; Merchant, M.; Moffat, J.; Schwarz, J. B. Discovery of (S)-1-(1-(4-chloro-3-fluorophenyl)-2-hydroxyethyl)-4-(2- ((1-methyl-1H-pyrazol-5-yl)-amino)pyrimidin-4-yl)pyridin-2(1H)-one (GDC-0994), an extracellular signal-regulated kinase 1/2 (ERK1/2) inhibitor in early clinical development. *J. Med. Chem.* **2016**, *59*, 5650−5660.

(4) For selected examples, see: (a) Greve, E.; Lindeman, S. V.; Scartelli, C.; Lin, L.; Flaumenhaft, R.; Dockendorff, C. Route exploration and synthesis of the reported pyridone-based PDI inhibitor STK076545. *Org. Biomol. Chem.* **2020**, *18*, 6665−6681. (b) Chorell, E.; Pinkner, J. S.; Phan, G.; Edvinsson, S.; Buelens, F.; Remaut, H.; Waksman, G.; Hultgren, S. J.; Almqvist, F. Design and Synthesis of C-2 Substituted Thiazolo and Dihydrothiazolo Ring-Fused 2-Pyridones: Pilicides with Increased Antivirulence Activity. *J. Med. Chem.* **2010**, *53*, 5690−5695. (c) Pfefferkorn, J. A.; Lou, J.; Minich, M. L.; Filipski, K. J.; He, M.; Zhou, R.; Ahmed, S.; Benbow, J.; Perez, A.-G.; Tu, M.; Litchfield, J.; Sharma, R.; Metzler, K.; Bourbonais, F.; Huang, C.; Beebe, D. A.; Oates, P. J. Pyridones as glucokinase activators: identification of a unique metabolic liability of the 4-sulfonyl-2-pyridone heterocycle. *Bioorg. Med. Chem. Lett.* **2009**, *19*, 3247−3252. (d) Verissimo, E.; Berry, N.; Gibbons, P.; Cristiano, M. L. S.; Rosenthal, P. J.; Gut, J.; Ward, S. A.; O'Neill, P. M. Design and synthesis of novel 2-pyridone peptidomimetic falcipain 2/3 inhibitors. *Bioorg. Med. Chem. Lett.* **2008**, *18*, 4210−4214. (e) Martinez, C. A.; Yazbeck, D. R.; Tao, J. An efficient enzymatic preparation of rhinovirus protease inhibitor intermediates. *Tetrahedron* **2004**, *60*, 759−764. (f) O'Connor, S.; Somers, P. J. Methods for the detection and quantitation of angiotensin converting enzyme inhibitors in fermentation broths. *J. Antibiot.* **1985**, *38*, 993−996.

(5) For selected examples, see: (a) Linghu, X.; Wong, N.; Iding, H.; Jost, V.; Zhang, H.; Koenig, S. G.; Sowell, G.; Gosselin, F. Development of a Practical Synthesis of ERK Inhibitor GDC-0994. *Org. Process Res. Dev.* **2017**, *21*, 387−398. (b) Yu, Y.; Zhu, C.; Wang, S.; Song, W.; Yang, Y.; Shi, J. Homosecoiridoid Alkaloids with Amino Acid Units from the Flower Buds of Lonicera japonica. *J. Nat. Prod.* **2013**, *76*, 2226−2233. (c) Minich, M. L.; Watson, I. D.; Filipski, K. J.; Pfefferkorn, J. A. Novel and efficient synthesis of 4-sulfonyl-2-pyridones. *Tetrahedron Lett.* **2009**, *50*, 2094−2096. (d) Martinez, C. A.; Yazbeck, D. R.; Tao, J. An efficient enzymatic preparation of rhinovirus protease inhibitor intermediates. *Tetrahedron* **2004**, *60*, 759−764.

(6) Rodrigues, A.; Lee, E. E.; Batey, R. A. Enantioselective Palladium(II)-Catalyzed Formal [3,3]-Sigmatropic Rearrangement of 2-Allyloxypyridines and Related Heterocycles. *Org. Lett.* **2010**, *12*, 260−263.

(7) (a) Li, C.; Kähny, M.; Breit, B. Rhodium-Catalyzed Chemo-, Regio-, and Enantioselective Addition of 2-Pyridones to Terminal Allenes. *Angew. Chem., Int. Ed.* **2014**, *53*, 13780−13784. (b) Li, C.; Breit, B. Rhodium-Catalyzed Dynamic Kinetic Asymmetric Allylation of Phenols and 2-Hydroxypyridines. *Chem. - Eur. J.* **2016**, *22*, 14655−14663.

(8) Zhang, X.; Yang, Z.-P.; Huang, L.; You, S.-L. Highly Regio-and Enantioselective Synthesis of *N*-Substituted 2-Pyridones: Iridium-Catalyzed Intermolecular Asymmetric Allylic Amination. *Angew. Chem., Int. Ed.* **2015**, *54*, 1873−1876.

(9) Khan, S.; Shah, B. H.; Khan, I.; Li, M.; Zhang, Y. J. Pd-Catalyzed regio- and enantioselective allylic substitution with 2-pyridones. *Chem. Commun.* **2019**, *55*, 13168−13171.

(10) Shimizu, I.; Tsuji, J. Palladium-catalyzed synthesis of 2, 3-disubstituted allylamines by regioselective aminophenylation or aminoalkenylation of 1, 2-dienes. *Chem. Lett.* **1984**, *13*, 233−236.

(11) For selected reviews, see: (a) Lechel, T.; Pfrengle, F.; Reissig, H.-U.; Zimmer, R. Three Carbons for Complexity! Recent Developments of Palladium-Catalyzed Reactions of Allenes. *ChemCatChem* **2013**, *5*, 2100−2130. (b) Jeganmohan, M.; Cheng, C.-H. Transition metal-catalyzed three-component coupling of allenes and the related allylation reactions. *Chem. Commun.* **2008**, *27*, 3101−3117. (c) Zimmer, R.; Dinesh, C. U.; Nandanan, E.; Khan, F. A. Palladium-catalyzed reactions of allenes. *Chem. Rev.* **2000**, *100*, 3067−3126.

(12) For selected reviews of Pd-catalyzed allylic substitution, see: (a) Lu, Z.; Ma, S. Metal-catalyzed enantioselective allylation in asymmetric synthesis. *Angew. Chem., Int. Ed.* **2008**, *47*, 258−297. (b) Trost, B. M.; Crawley, M. L. Asymmetric transition-metal-catalyzed allylic alkylations: applications in total synthesis. *Chem. Rev.* **2003**, *103*, 2921−2944.

(13) For Pd-catalyzed asymmetric allylic alkylation of allenes, see: (a) Blieck, R.; Taillefer, M.; Monnier, F. Metal-Catalyzed Intermolecular Hydrofunctionalization of Allenes: Easy Access to Allylic Structures via the Selective Formation of C−N, C−C, and C−O Bonds. *Chem. Rev.* **2020**, *120*, 13545−13598. For selected examples, see: (b) Zhu, M.; Zhang, Q.; Zi, W. Diastereodivergent Synthesis of *β*-Amino Alcohols by Dual-Metal Catalyzed Coupling of Alkoxyallenes with Aldimine Esters. *Angew. Chem., Int. Ed.* **2021**, *60*, 6545−6552. (c) Hilpert, L. J.; Sieger, S. V.; Haydl, A. M.; Breit, B. Palladium- and Rhodium-Catalyzed Dynamic Kinetic Resolution of Racemic Internal Allenes Towards Chiral Pyrazoles. *Angew. Chem., Int. Ed.* **2019**, *58*, 3378−3381. (d) Zhou, H.; Wang, Y.; Zhang, L.; Cai, M.; Luo, S. Enantioselective Terminal Addition to Allenes by Dual Chiral Primary Amine/Palladium Catalysis. *J. Am. Chem. Soc.* **2017**, *139*, 3631−3634. (e) Liu, J.; Han, Z.; Wang, X.; Wang, Z.; Ding, K. Highly Regio- and Enantioselective Alkoxycarbonylative Amination of Terminal Allenes Catalyzed by a Spiroketal-Based

https://doi.org/10.1021/acs.orglett.1c00959
Org. Lett. 2021, 23, 3567−3572

Diphosphine/Pd(II) Complex. *J. Am. Chem. Soc.* **2015**, *137*, 15346−15349. (f) Lim, W.; Kim, J.; Rhee, Y. H. Pd-Catalyzed Asymmetric Intermolecular Hydroalkoxylation of Allene: An Entry to Cyclic Acetals with Activating Group-Free and Flexible Anomeric Control. *J. Am. Chem. Soc.* **2014**, *136*, 13618−13621. (g) Trost, B. M.; Jäkel, C.; Plietker, B. Palladium-Catalyzed Asymmetric Addition of Pronucleophiles to Allenes. *J. Am. Chem. Soc.* **2003**, *125*, 4438−4439.

(14) Only two examples have been reported in favor of branched selectivity for the three-component carbofunctionalization of allenes, see: (a) Li, H.; Li, T.; Hsueh, Y. J.; Wu, X.; Xu, F.; Zhang, Y. J. Tandem arylation and regioselective allylic etherification of 2, 3-allenol via Pd/B cooperative catalysis. *Org. Biomol. Chem.* **2019**, *17*, 8075−8078. (b) Ma, S.; Zhao, S. Novel substituent and chelating effects in the Pd-catalyzed reaction of 2,3-allenols, aryl iodides, and amines. Highly regio-and stereoselective synthesis of 2-amino-3-alken-1-ols or 4-amino-2-(*E*)-alken-1-ols. *J. Am. Chem. Soc.* **2001**, *123*, 5578−5579.

(15) (a) Khan, A.; Khan, S.; Khan, I.; Zhao, C.; Mao, Y.; Chen, Y.; Zhang, Y. J. Enantioselective Construction of Tertiary C−O Bond via Allylic Substitution of Vinylethylene Carbonates with Water and Alcohols. *J. Am. Chem. Soc.* **2017**, *139*, 10733−10741. (b) Khan, S.; Li, H.; Zhao, C.; Wu, X.; Zhang, Y. J. Asymmetric allylic etherification of vinylethylene carbonates with diols via Pd/B cooperative catalysis: A route to chiral hemi-crown ethers. *Org. Lett.* **2019**, *21*, 9457−9462.

(16) (a) Khan, A.; Zhang, Y. J. Palladium-Catalyzed Asymmetric Decarboxylative Cycloaddition of Vinylethylene Carbonates with Electrophiles: Construction of Quaternary Stereocenters. *Synlett* **2015**, *26*, 853−860. (b) Zhao, C.; Khan, I.; Zhang, Y. J. Enantioselective total synthesis of furofuran lignans via Pd-catalyzed asymmetric allylic cycloadditon of vinylethylene carbonates with 2-nitroacrylates. *Chem. Commun.* **2020**, *56*, 12431−12434. (c) Zhao, C.; Shah, B. H.; Khan, I.; Kan, Y.; Zhang, Y. J. Enantioselective Synthesis of Isoxazoline *N*-Oxides via Pd-Catalyzed Asymmetric Allylic Cycloaddition of Nitro-Containing Allylic Carbonates. *Org. Lett.* **2019**, *21*, 9045−9049. (d) Khan, I.; Shah, B. H.; Zhao, C.; Xu, F.; Zhang, Y. J. Pd-Catalyzed Asymmetric Allylic Cycloaddition of *N*-Containing Allylic Carbonates with Isocyanates. *Org. Lett.* **2019**, *21*, 9452−9456. (e) Xu, H.; Khan, S.; Li, H.; Wu, X.; Zhang, Y. J. Pd-Catalyzed Asymmetric Allylic Cycloaddition of Vinyloxetanes with Formaldehyde. *Org. Lett.* **2019**, *21*, 214−217. (f) Liu, K.; Khan, I.; Cheng, J.; Hsueh, Y. J.; Zhang, Y. J. Asymmetric decarboxylative cycloaddition of vinylethylene carbonates with *β*-nitroolefins by cooperative catalysis of palladium complex and squaramide. *ACS Catal.* **2018**, *8*, 11600−11604. (g) Khan, A.; Xing, J.; Zhao, J.; Kan, Y.; Zhang, W.; Zhang, Y. J. Palladium-Catalyzed Enantioselective Decarboxylative Cycloaddition of Vinylethylene Carbonates with Isocyanates. *Chem. - Eur. J.* **2015**, *21*, 120−124. (h) Khan, A.; Zheng, R.; Kan, Y.; Ye, J.; Xing, J.; Zhang, Y. J. Palladium-Catalyzed Decarboxylative Cycloaddition of Vinylethylene Carbonates with Formaldehyde: Enantioselective Construction of Tertiary Vinylglycols. *Angew. Chem., Int. Ed.* **2014**, *53*, 6439−6442. (i) Khan, A.; Yang, L.; Xu, J.; Jin, L. Y.; Zhang, Y. J. Palladium-Catalyzed Asymmetric Decarboxylative Cycloaddition of Vinylethylene Carbonates with Michael Acceptors: Construction of Vicinal Quaternary Stereocenters. *Angew. Chem., Int. Ed.* **2014**, *53*, 11257−11260.

(17) (a) Wang, X.; Han, Z.; Wang, Z.; Ding, K. A Type of Structurally Adaptable Aromatic Spiroketal Based Chiral Diphosphine Ligands in Asymmetric Catalysis. *Acc. Chem. Res.* **2021**, *54*, 668−684. (b) Wang, X.; Meng, F.; Wang, Y.; Han, Z.; Chen, Y.-J.; Liu, L.; Wang, Z.; Ding, K. Aromatic Spiroketal Bisphosphine Ligands: Palladium-Catalyzed Asymmetric Allylic Amination of Racemic Morita-Baylis-Hillman Adducts. *Angew. Chem., Int. Ed.* **2012**, *51*, 9276−9282. (c) Wang, X.; Han, Z.; Wang, Z.; Ding, K. Catalytic Asymmetric Synthesis of Aromatic Spiroketals by SpinPhox/Iridium (I)-Catalyzed Hydrogenation and Spiroketalization of *α*,*α*′-Bis(2-hydroxyarylidene) Ketones. *Angew. Chem., Int. Ed.* **2012**, *51*, 936−940.

(18) Zhou, Q.; Zheng, L.; Ma, B.; Huang, L.; Liu, A.; Cao, X.; Yu, J.; Ma, X. Insights into Substrate Self-Assisted Activation of Allylic Alcohols Guiding to Mild Allylic Substitution of Tautomerizable Heteroarenes. *J. Org. Chem.* **2020**, *85*, 5097−5103.

(19) For a review of Prins cyclization, see: Olier, C.; Kaafarani, M.; Gastaldi, S.; Bertrand, M. P. Synthesis of tetrahydropyrans and related heterocycles via prins cyclization; extension to aza-prins cyclization. *Tetrahedron* **2010**, *66*, 413−445.

(20) Babaev, E. V.; Koval, Y. I.; Rybakov, V. B. Dehydrohalogenation of isomeric 2- and 3-bromomethyl substituted 2,3-dihydrooxazolo-[3,2-a]pyridines. *Mendeleev Commun.* **2020**, *30*, 228−230.

https://doi.org/10.1021/acs.orglett.1c00959
Org. Lett. 2021, 23, 3567−3572